RETRAINING THE UPPER LIMB

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PT7003/PT4000

LEARNING OUTCOMES

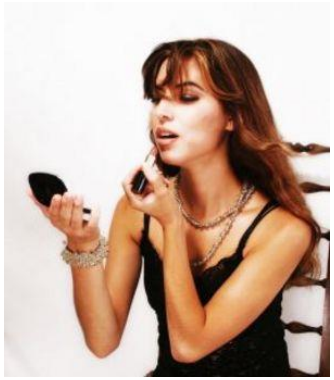
After this lecture the student should be able to:

1. Discuss the key elements of normal UL movement
2. Describe common problems of the upper limb following neurological problems
3. Identify common interventions used for upper limb rehabilitation

FUNCTIONS OF THE UPPER LIMB





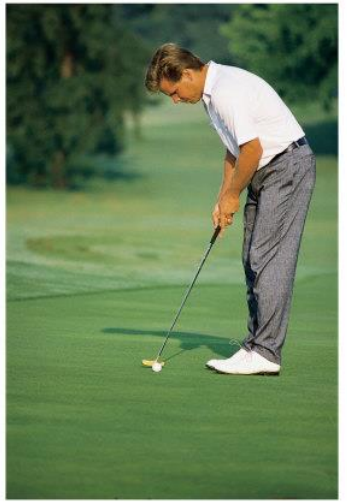
















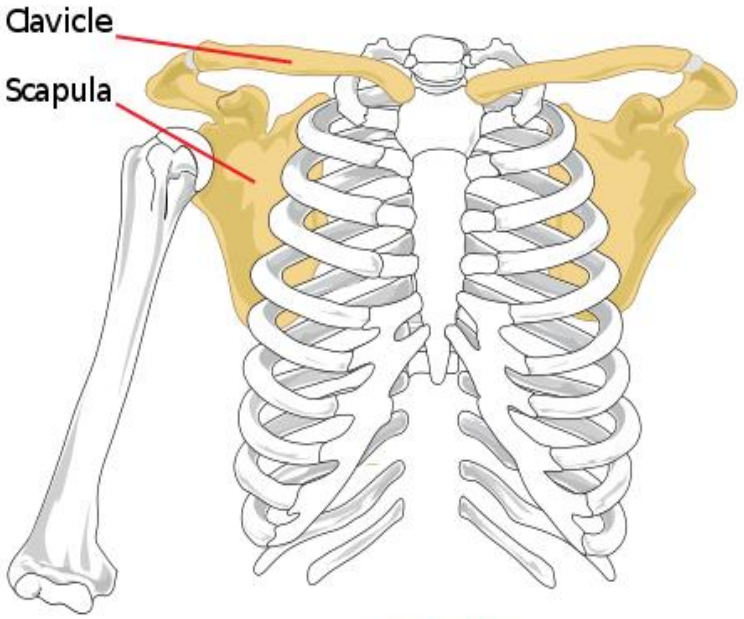
ANATOMY THAT SUPPORT UPPER LIMB FUNCTION (PECTORAL GIRDLE)

Scapula (2)

Clavicle (2)

Function

- Stable base for upper limb function
- Facilitate upper limb movement



•Kinetic chain

- UL function reliant on optimal function of other segments e.g. trunk, cervical, lower limb etc

e.g. Proximal control

- Pelvis & pectoral girdles

KEY COMPONENTS OF UPPER LIMB CONTROL

Two components (Jeannerod 1984)

Transportation phase

Manipulation

Four components (Shumway-Cook & Woollacott 2012)

Visual regard

Transportation phase

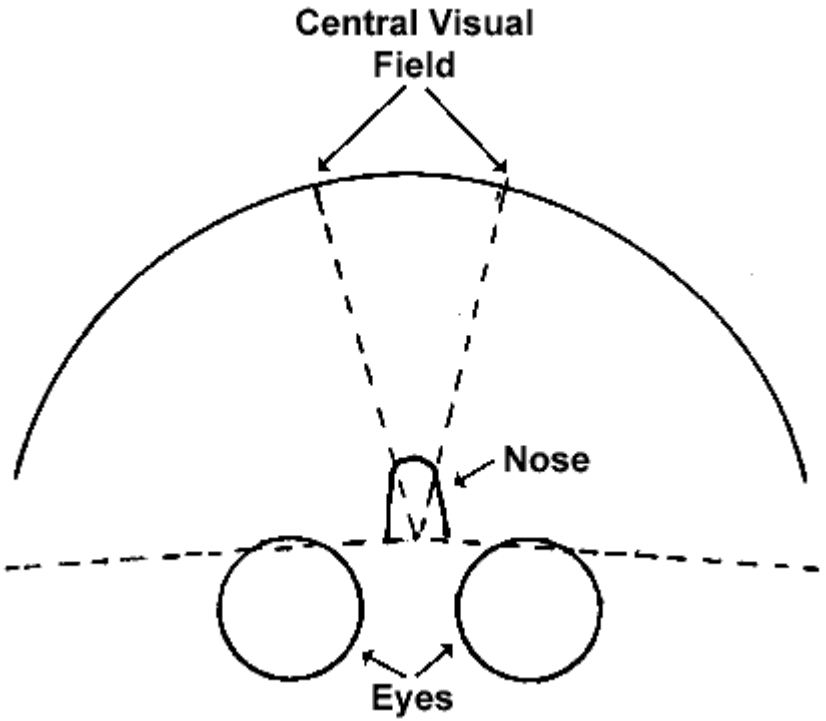
Grip formation phase

In-hand manipulation phase

LOCATING A TARGET (VISUAL REGARD)

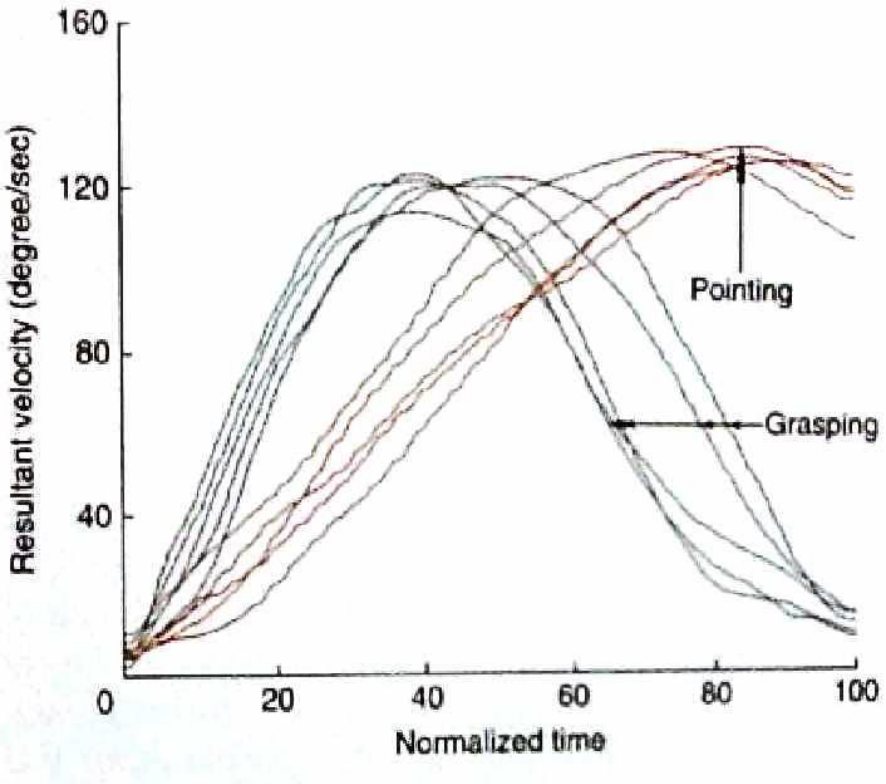
Vision

- Eyes alone (central field)
- Eyes and head (near peripheral visual field)
- Eyes, head and trunk (far periphery)



Reaching

TRANSPORTATION PHASE - KINEMATICS



Initial acceleration & prolonged deceleration phase

The temporal characteristic is determined by the task

Total transport time & relative timing of phases

- Goal (grasp, lift, point, ballistic movement)
- Task constraints (Grasp to fit vs grasp to throw, grasp to drink etc)
- Speed accuracy trade-off

Relevance for rehabilitation

Reaching

GRASPING & MANIPULATION

Hand is the principal means to interact with people & objects in the external environment

Anatomical structure & specific cortico-motoneuronal connections to hand muscles allow many combinations of joint rotations and movement possibilities

Approach to grasp and manipulation is determined by:

- Intention
- Object characteristics
- Task constraints

Studies have found that when the fingers do manipulative tasks, the pattern is one of:

- Fractionating of muscle activity

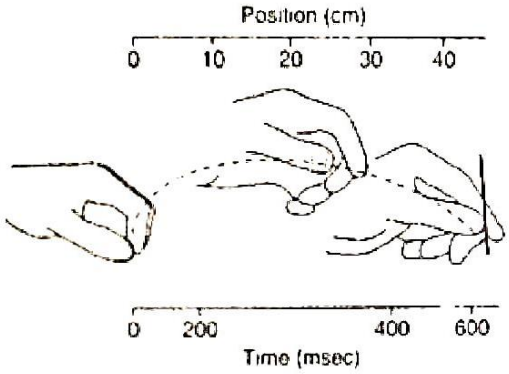
During gripping tasks

- A pattern of co-contraction emerges with the number of muscles involved depending on the force exerted

GRASPING & MANIPULATION - KINEMATICS

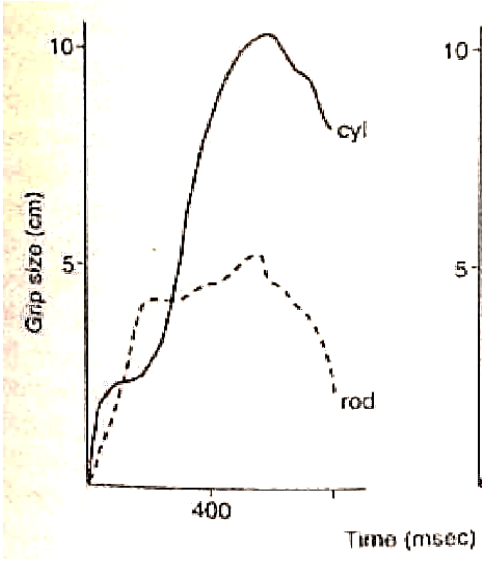
Grip formation

- with transport phase
- Anticipatory
- grip opening achieved with finger movements
- reduction to match object size under visual guidance



Pre-grasp hand shaping

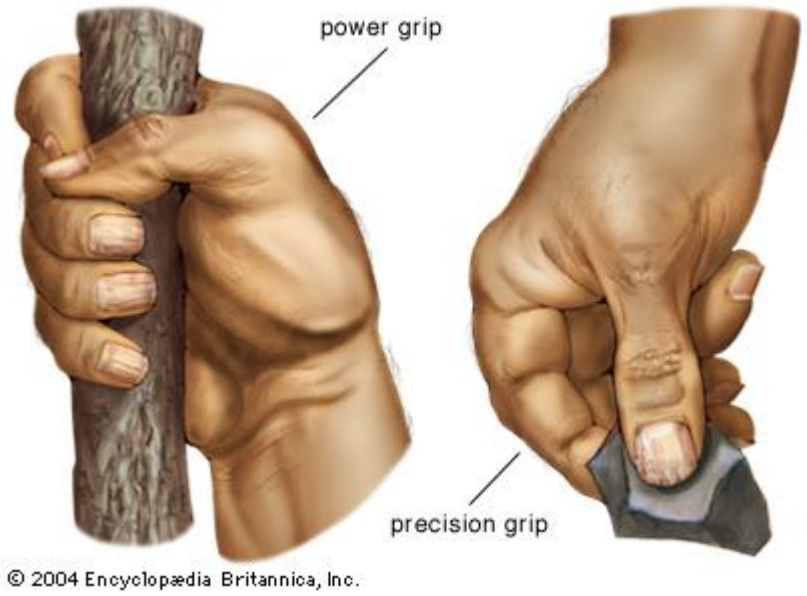
- Intrinsic object properties (shape, size, texture)
- Extrinsic object properties (orientation, distance from body, location with respect to body)
- Intended use



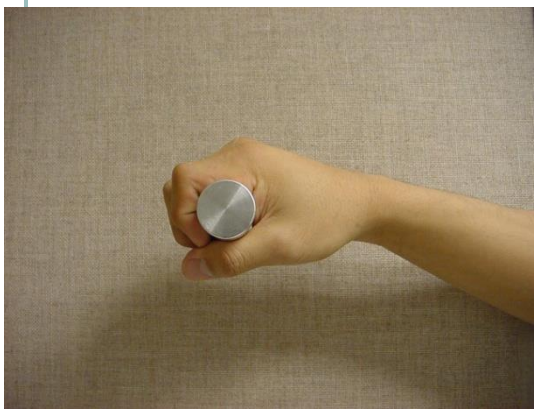
GRIPING – POWER AND PRECISION

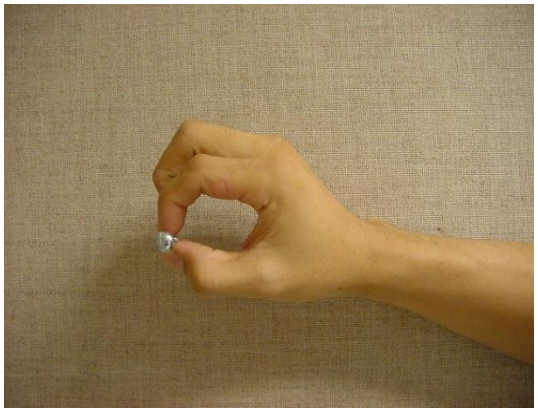
Napier classification

- Precision and power grasp

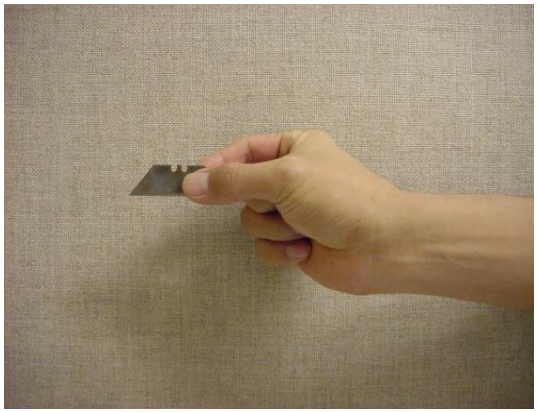


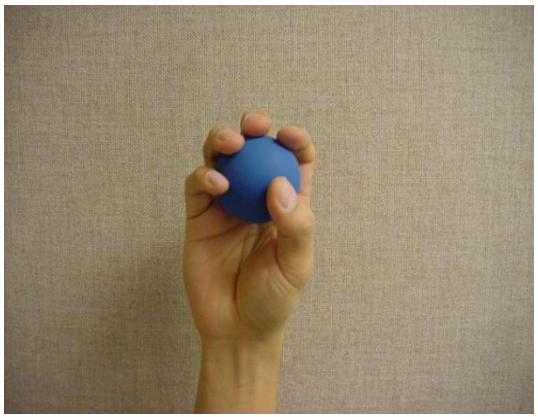
TYPES OF GRIPS IN PREHENSION

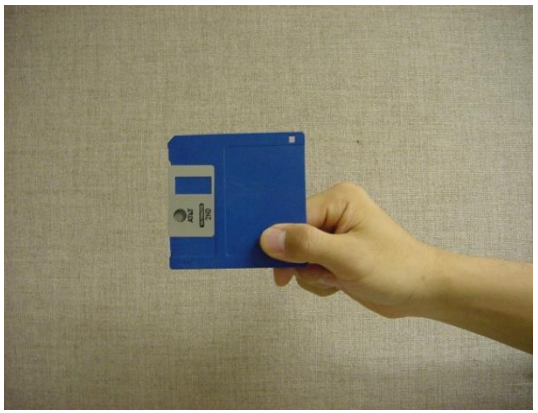












reaching

IN HAND MANIPULATION SKILLS

Translation

Rotation

Shift



TRUNK, LOWER LIMB AND ARM MOVEMENT

Reaching out in standing

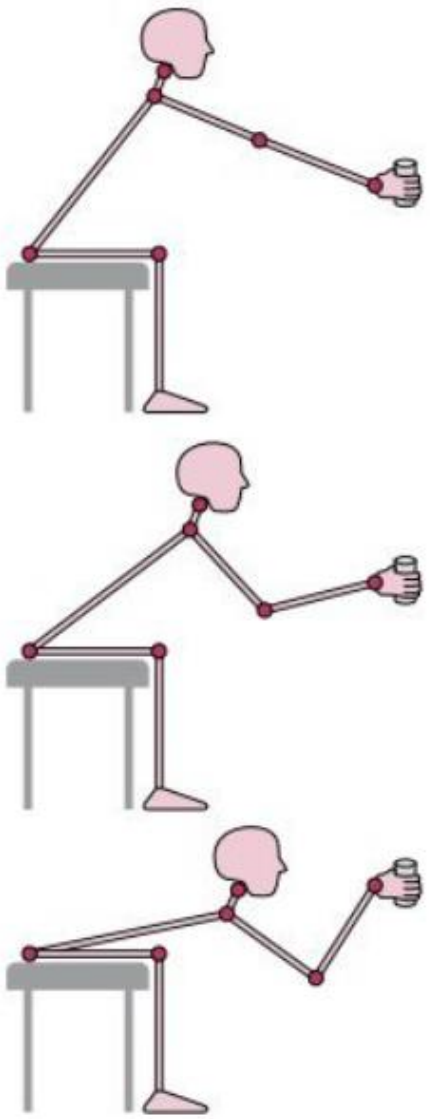
- Preparatory postural adjustments
- Appropriate temporal co-ordination between leg and arm muscles
- Studies of stroke survivors have shown force & EMG activities in gastrocnemius

Reaching to different part of workspace

- Within arms length: shoulder, elbow, wrist, forearm, and hand joints
- Beyond arms length: shoulder, elbow, wrist, forearm, and hand joints + varying degrees of trunk activities
- The hip, shoulder girdle and glenohumeral joints functions as a multi-segmental linkage

Reaching and throwing tasks

- Involves varying contributions from different segments
- In each case, proximal activations preceded arm activation
- Requirements of postural stability affects the accuracy and speed of the tasks



BIMANUAL REACHING TASKS

Upper limbs co-operate together to perform the task

- E.g. anticipatory postural adjustments during ball drop into a cup held by the other hand
- Speed of hand movement is affected when both hands are used (if one hand performs a more complex task requiring longer time, the other hand slows down)
- Given the complexity of bi-manual tasks, it is necessary to provide opportunities for the patient to practice such tasks otherwise regaining this will not occur

Upper limb in weight bearing

- Tasks such as pushing a drawer shut, push the door, or push oneself up the floor (Closed-chain activities)
- Movement occurs principally at the shoulder and elbow
- Shoulder and elbow extensor muscles (Triceps brachii)
- Flexible wrist to allow necessary extension so that pushing can occur through palm of hand

VIDEO 1



COMMON UL PROBLEMS FOLLOWING NEUROLOGICAL DEFICITS

Muscle weakness

Sensory deficits (visual, somatosensory)

Altered tone –spasticity/ hypotonicity

Motor performance issues

- Poor timing and coordination of segmental movements
- Inter-joint co-ordination issues
- Atypical movement patterns e.g., Synergies

Other secondary impairments e.g. muscle length changes, contractures etc

Perceptual deficits such as apraxia, unilateral neglect

MUSCLE WEAKNESS

GH joint abductors, flexors, and external rotators and elbow supinators - affects reaching actions

Wrist extensors, finger and thumb flexors, extensors, abductors, and adductors: affect manipulation of objects

Difficulties sustaining and controlling grip force during grasping and lifting objects is also a common problem

Trunk weakness also common (Dickstein et al. 2004)

Ipsilateral upper limb weakness also common 9Newham & Hsaio 2001)

SENSORY DEFICITS

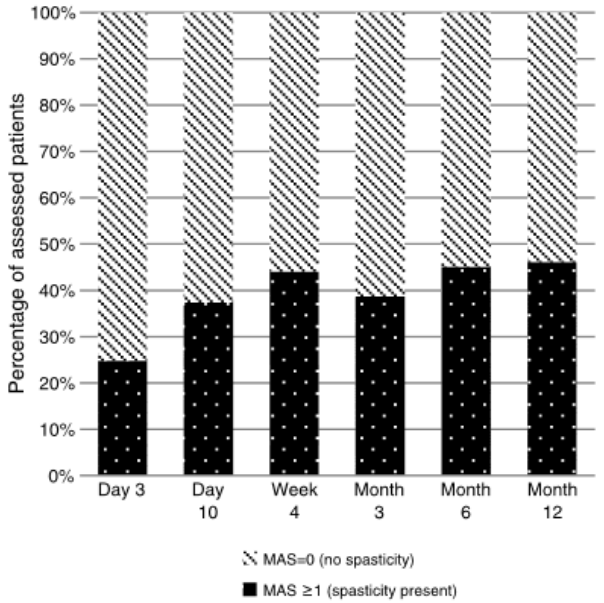
Touch	53% - tactile sensations
Proprioception	89% - impaired stereognosis,
Visual deficits	64% - impaired proprioception.
Learned non-use	

Problems

SPASTICITY

Prevalence

- 19% - first 3 months
- 27-43% (of upper limb paresis) in 18 months
- 4-13% disabled by it
- Associated with pain & limited ROM
- Impacts functional ability



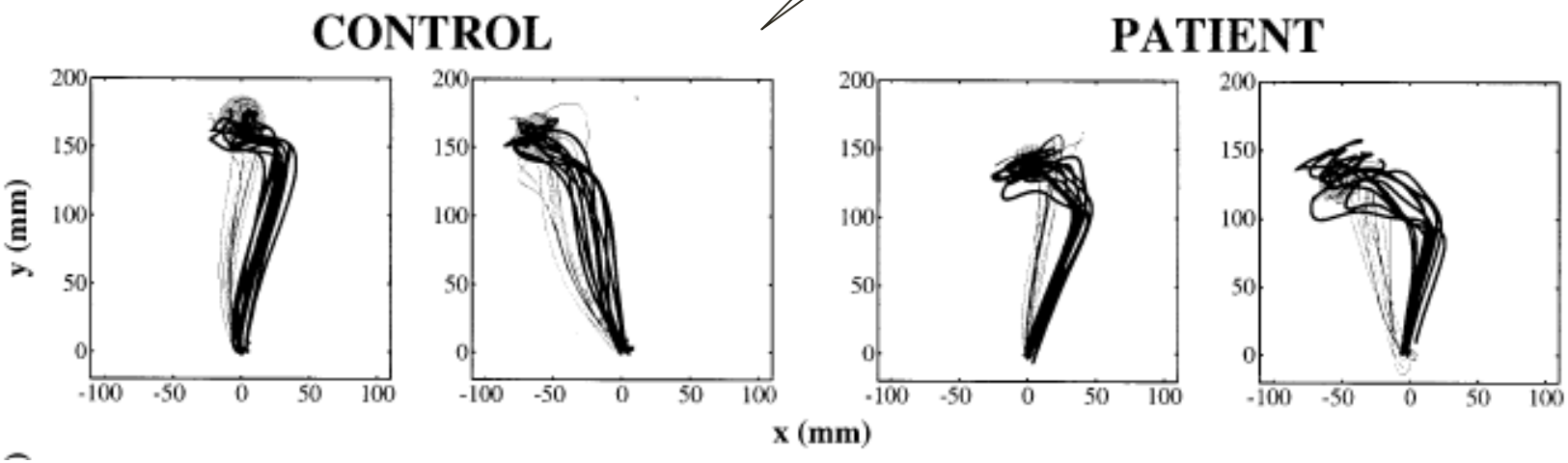
Pain, limited ROM and loss of sensation more frequent in those with spasticity at week 3 and 12 months.

Sommerfield DK et al. 2012

Opheim A et al. 2014

POOR TIMING - REACHING

- Slower movement times
- Less accurate
- Undershooting
- Overshooting
- Decomposition



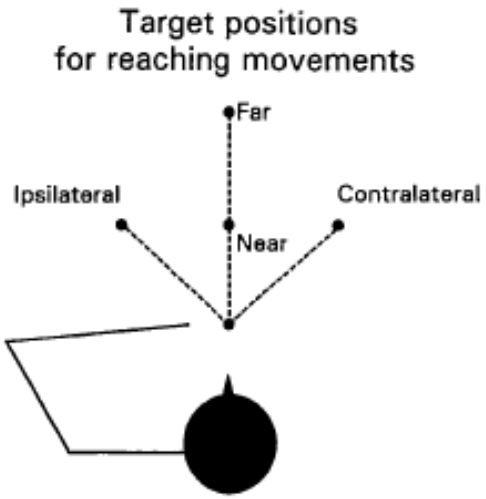
Cirstea MC et al. 2003

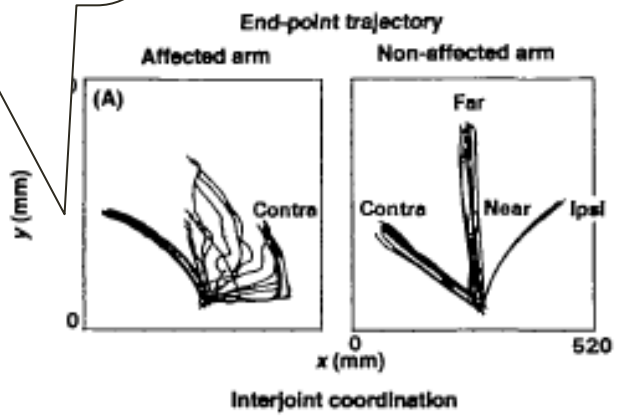
INTERJOINT CO-ORDINATION

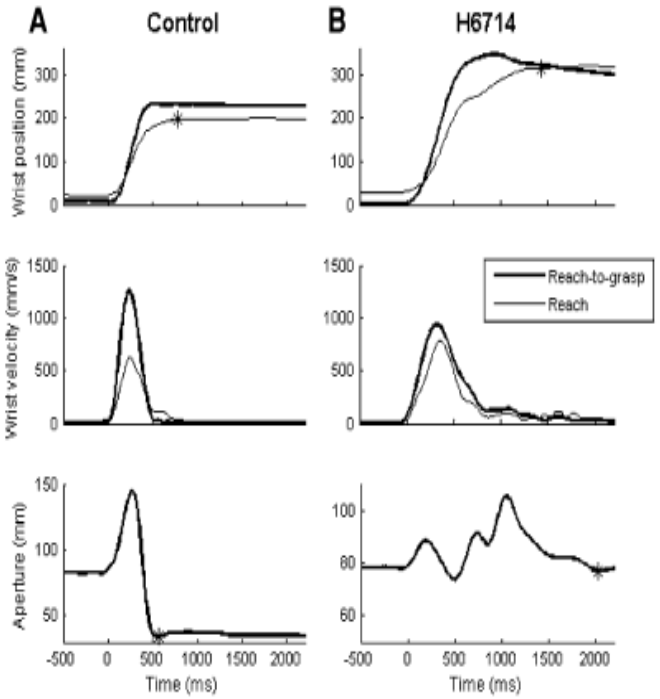
Deficits in reaching

↓ movt velocity
Earlier aperture
Difficulty achieving the required aperture required

Longer movement times
Smaller amplitudes
Segmentation
Variability
Inter-joint in-cordination







Levin MF (1996)

Problems

ADAPTIVE MOVEMENT STRATEGIES

Reaching

- Elevation of shoulder girdle
- Lateral flexion and rotation of spine



Manipulation

- Excessive hand opening
- Excessive grasping force



Problems

MUSCLE LENGTH CHANGES/CONTRACTURE

Immobility

Disuse

Adaptive movement patterns often reflects

- paresis, imbalance and soft tissues inflexibilities

Muscles typically affected by contracture are:

- **Shoulder adductors, internal rotators, elbow flexors, pronators, wrist and finger flexors and thumb adductors**

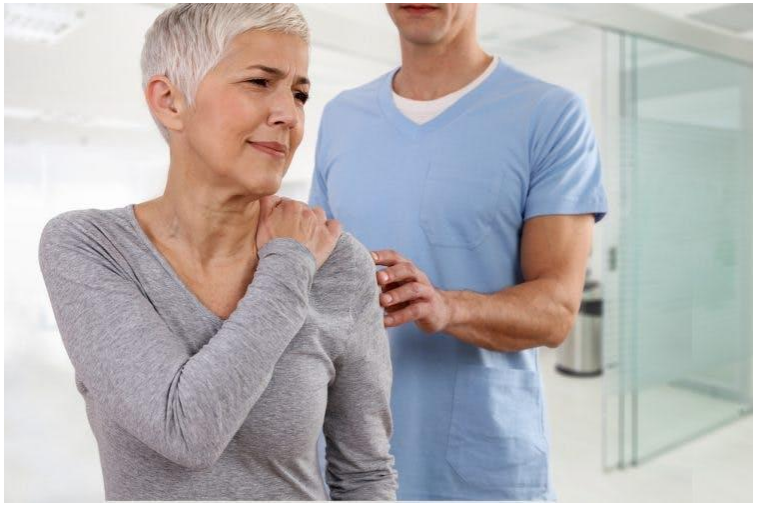
Hypersensitivity may develop following shortening and stiffening of the muscle - spasticity

Muscle activation and active use must be prioritised early



COMMON COMPLICATIONS

Hemiplegic shoulder pain



Shoulder subluxation



APRAXIA

Inability to carry out purposeful movement in the presence of intact motor, sensory and co-ordination abilities

This is largely due to the damage to the area of the brain responsible for motor planning

Ideomotor apraxia

- Unable to execute or imitate or produce a purposeful motor task on command but can occur automatically

Ideational (frontal) apraxia

- Not possible on command or automatically

problems

EVIDENCE BASED UPPER LIMB REHABILITATION

Isaac Sorinola

ASSESSMENT

Functional assessment

- **BADL**
  - Barthel Index
  - FIM
- **IADL**
  - Telephone
  - Travelling
  - Shopping
  - Preparing meals /Housework
  - Finances
- **Arm & Hand function**
  - ARAT
  - JTHFT
  - Motoricity Index
  - Nine-hole peg Test

Strategy assessment

- Reaching & grasp
- Grasp formation
- Manipulation & release

Impairments assessment

- Perception & cognition
- Musculoskeletal & neuromuscular factors
  - ROM, strength, oedema, sensation, pain

http://www.rehabmeasures.org/default.aspx

TREATMENT PLANNING

Long term goals

- Expressed in terms of:
  - Functioning - self-care, work or leisure activities.
    - E.g. Be able to prepare a morning meal or groom, write or use the computer independently

Short terms goals

- Expressed in terms of:
  - Resolving impairments,
  - Deficits in strategies
  - Interim steps to achieve long term goals.
    - E.g. improve reach distance etc.

Both should be client centred and SMART

Topics

RETRAINING REACHING AND MANIPULATION

Effective functional use of the hand - dependent on functional hand grasp and release

Movement of the upper limb - driven by the hand and the placement of the hand in a person's close environment or workspace

Consequently, reaching practice should always involve:

- Use of a target or an object in hand
- Intensive & Repetitive practice of finger and wrist exercises
- Early initiation of practice
- With severe weakness, emphasis is on eliciting activities in key muscles plus repetitive practice of simple movements
- Reaching and finger movements involved in grasp and release are emphasised
- Focus - to promote activity for different tasks and prevent contractures in at-risks muscles

Apply motor learning principles: instruction, feedback, external focus etc

Task modification: modify what is to be practiced, taking into account the individual's motor impairments

Constraint of body segment

PHYSIOTHERAPY MANAGEMENT – UPPER LIMB

Task Oriented training method

- Simple repetitive exercises
- Functional muscle strength and endurance training
- Bimanual practice
- Task training and practice plus constraint

Management of impairmen







- Muscle weakness
- Soft tissue length
- Sensory re-education
- Pain

Other Strategies

- Electrical stimulation–EMG triggered stimulation, passive stimulation

SIMPLE REPETITIVE EXERCISE

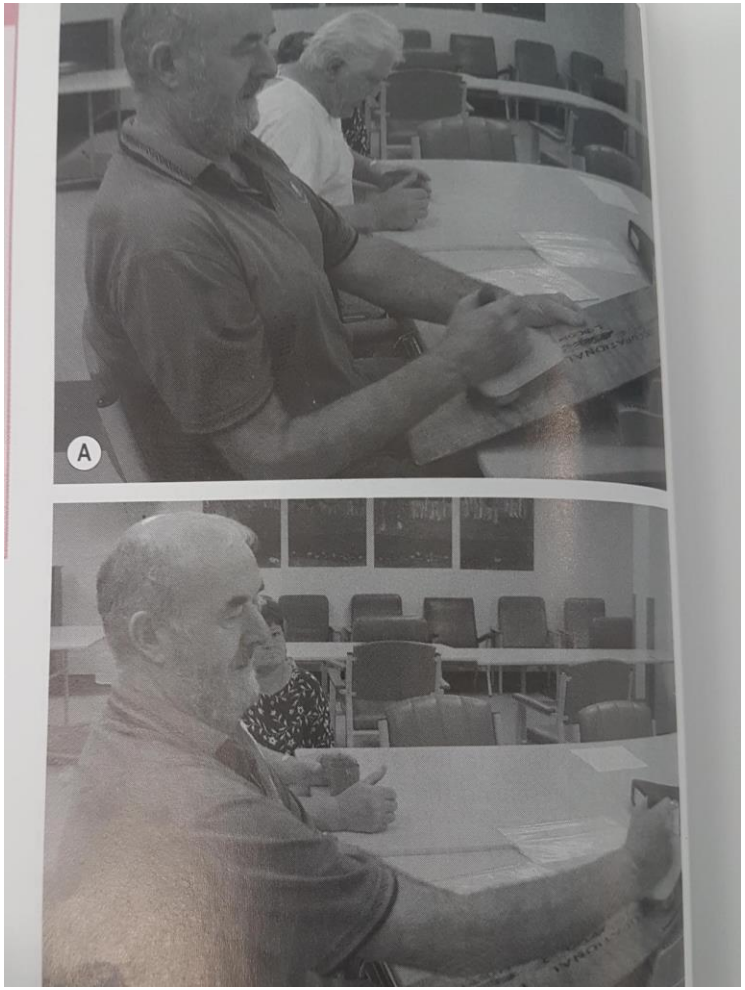
When there is an inability to control multi-segmental movement

Often directed at one muscle group

- Reaching forward (shoulder flexion, elbow extension)
- Shoulder external rotation plus arm supination, elbow flexed
- Wrist flexion and extension in mid-rotation of fore-arm
- Finger flexion/extension
- Grasping, releasing and placing different objects

EMG triggered stim, Electrical stimulation, non-robotic, or robotic trainer

Studies of repetitive practice of simple exercises have found beneficial functional improvements



FUNCTIONAL STRENGTH TRAINING

Studies have shown that functional strength training improves functional performance

Box 6.3 Strength training

- Gripping exercises using grip-force dynamometer, spring-resisted gripping device or plastic putty, pouring water from vessel to vessel (increasing weight of one, decreasing weight of the other); various types of grip are practised with different objects
- Use of progressively heavier objects in reaching, lifting and manipulating tasks
- Elastic band exercises, using progressively tighter bands
- Pushing, pulling exercises, using body weight or elastic bands
- Arm cycling (bimanual or unimanual)

BIMANUAL PRACTICE

Important because the Upper Limb work co-operatively in most everyday functions e.g.

- Unscrewing a jar
- Folding clothes
- Making tea
- Crying a tea cup etc

How:

- Early – as soon as patient have ability with simple movements
- Practice simple tasks in which both hands work co-operatively
  - Do a task where both hands are performing different action – e,g, opening a can
  - Do a task where both hands are doing same action e.g. folding a towel
  - Working in opposite directions using a gardening shears
  - Doing rhythmical sequential actions like arm cycling

















CONTRAINT INDUCED MOVEMENT THERAPY (CIMT)

Based on the work of Taub in the early 2000.

Box 6.4 Protocol for task-specific training plus constraint

Criteria for inclusion

- Wolf's Minimum Motor Criteria: ability to extend wrist at least 20°, interphalangeal and MCP finger joints at least 10°
- Willingness of patient to attend initially for a training programme 5 days per week for 2 weeks

Constraint

- Hand splint with wrist of non-paretic arm in some extension + sling, or mitt/glove + reminders not to use the non-paretic limb
- Wear splint for a prescribed number of hours on each week day for 2 weeks

Behavioural contract

- Seek agreement with patient about activities to be performed without constraint: e.g., bathing, parts of dressing, activities where balance may be an issue and, with constraint: e.g., eating, grooming, toileting, some domestic tasks

Programme of exercises

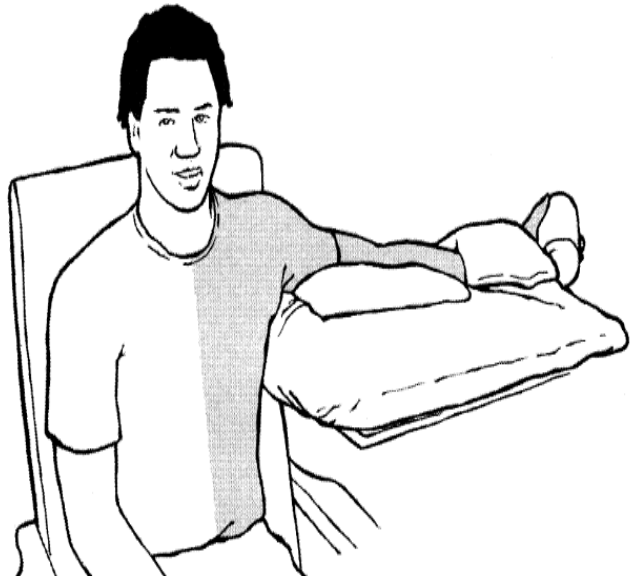
- Prescribed number of hours of task-oriented training and practice for the paretic limb + home practice

Measurement

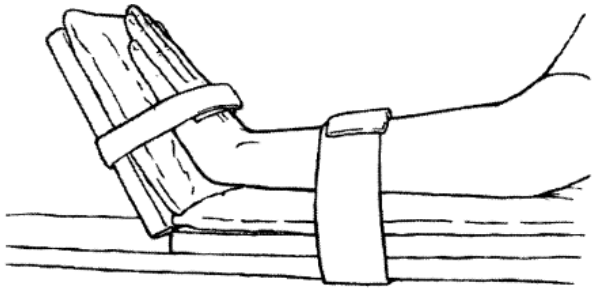
- Performed before and after 2-week programme and after a few weeks (for retention)



Inhibit abnormal tone and movements, maintain soft tissue length



Correct positioning







Stretches

- **Slow stretch** (Carr and Shepherd, 1998), 20 min daily

- **Brief passive stretches** carried out immediately before exercise session to decrease muscle stiffness and throughout exercises as needed

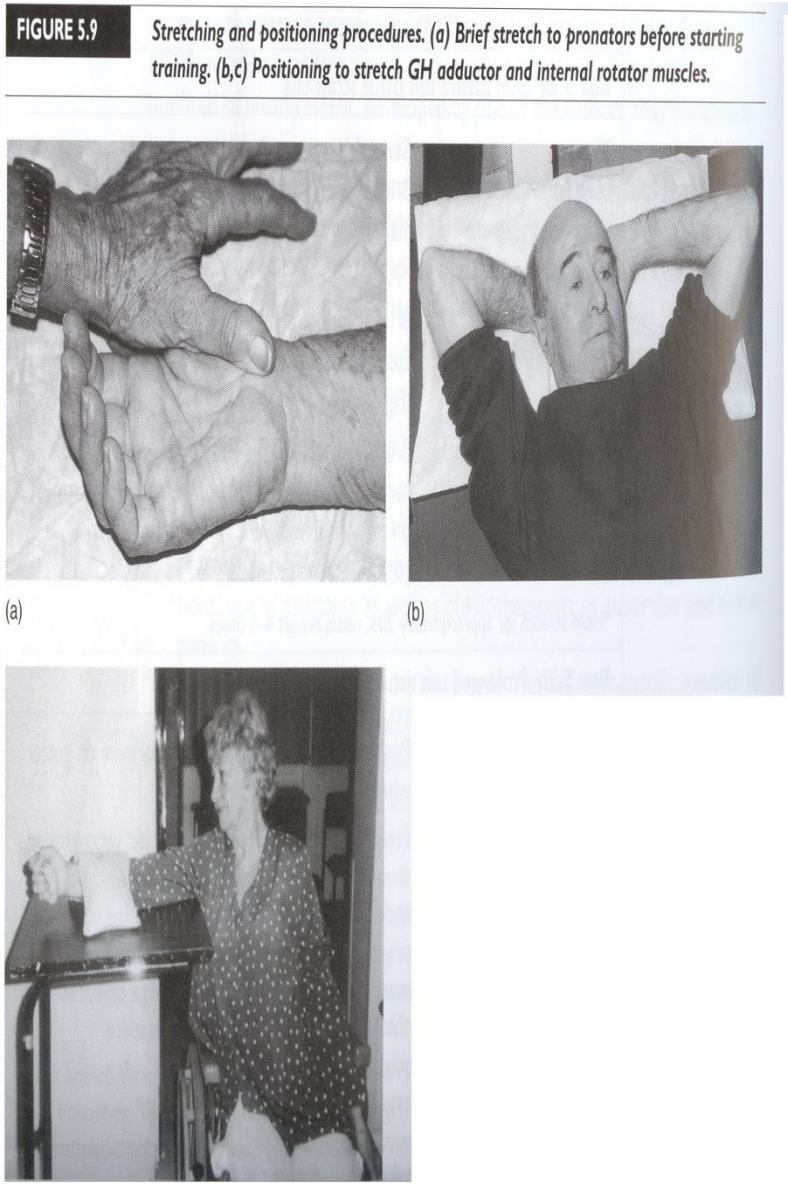


FIGURE 5.9 *Stretching and positioning procedures. (a) Brief stretch to pronators before starting training. (b,c) Positioning to stretch GH adductor and internal rotator muscles.*

(a)

(b)

SENSORY RE-EDUCATION

- Sensory re-education
  - Task-related training with the manipulation of objects with different characteristics for a variety of different purposes
  - Training conducted with vision and without vision





OTHER POSSIBLE STRATEGIES

EMG triggered stimulation

- No difference to usual therapy for strength and activity in early stroke (Dorsch et al. 2014) or chronic stroke (Singer BJ et al. 2013)

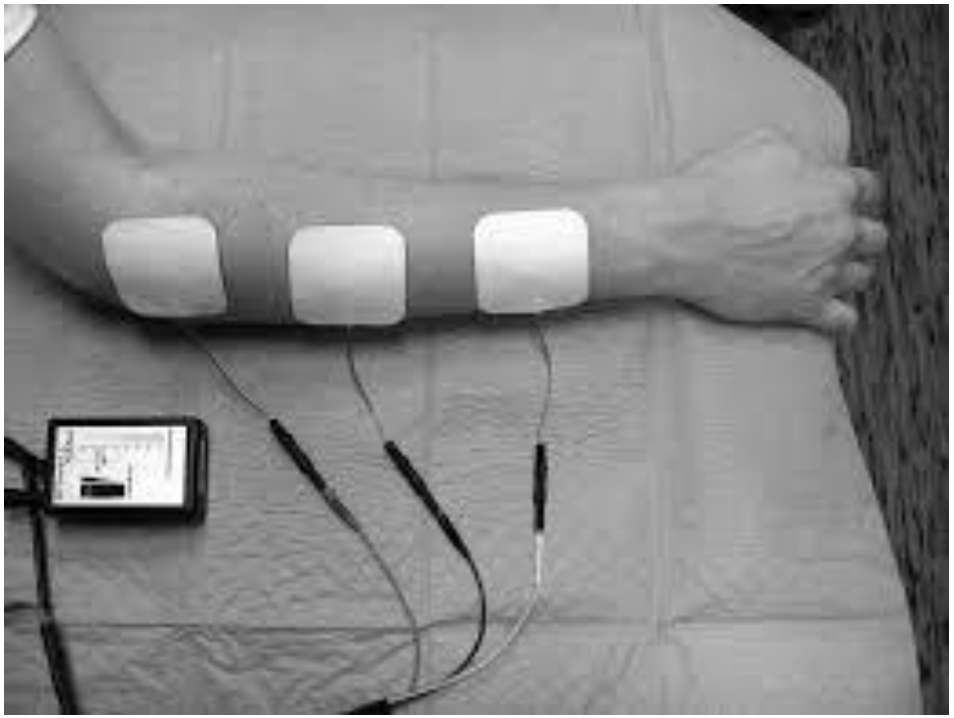
Mirror therapy

- Better FMA & FIM changes in mirror therapy group (Park et al. 2015; Yavuzer G et al. 2008; Samuelkamaleshkumar S et al. 2014)

Mental Imagery

- Effect size 6.75 for ARAT (Kho et al. 2014)

EMG TRIGGERED STIMULATION



MIRROR THERAPY



SAEBOFLEX

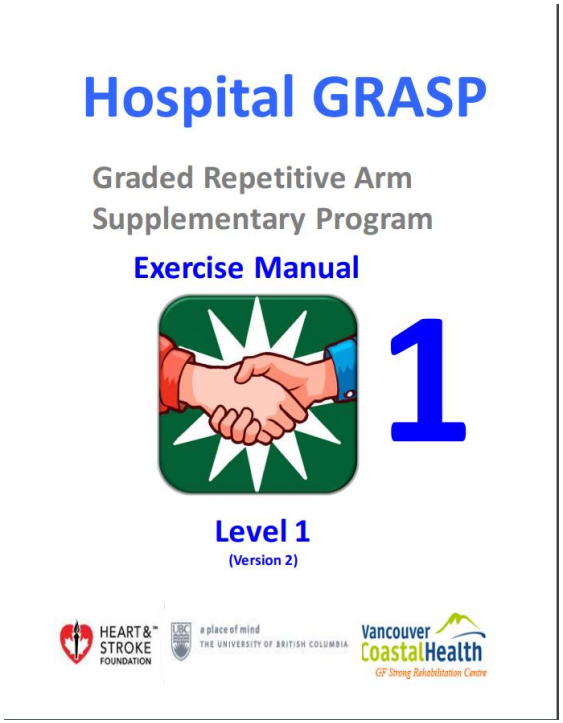


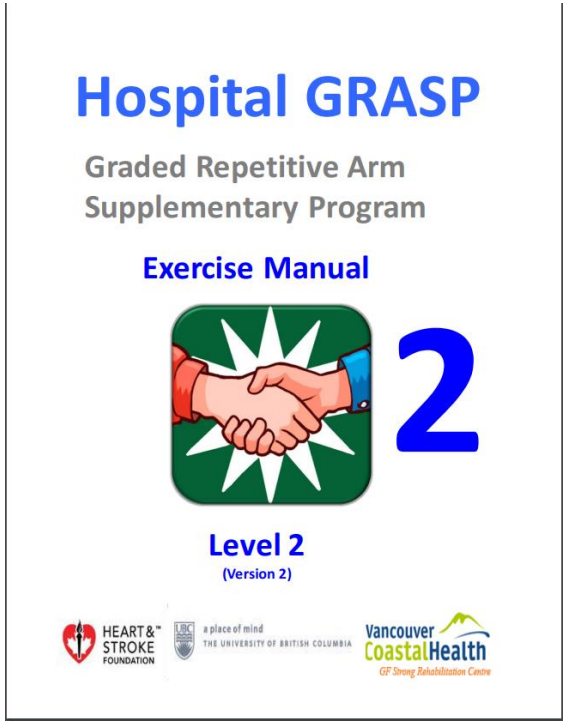


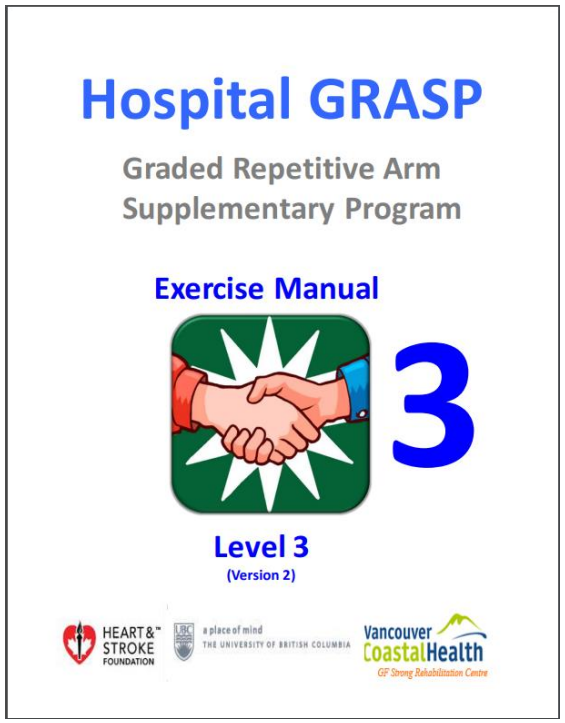


Evidence -Limited

Farrell et al. (2007) - ↑ movement at the shoulder and elbow.
↑ Wrist extension,
but wrist flexion and finger movement did not.
↑ FMA & MAS
↓Summed Ashworth (spasticity)







https://neurorehab.med.ubc.ca/grasp/

GRASP EXERCISES (GRADED REPETITIVE ARM SUPPLEMENTARY PROGRAM)

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